Quantum Key Distribution with BB84 and BBM92

Simulation study of intercept-and-resend attacks, Pauli channel noise, and sample-based QBER estimation

Abstract. This project implements educational simulations of the BB84 prepare-and-measure protocol and the entanglement-based BBM92 protocol using Qiskit. Intercept-and-resend attacks are evaluated at six attack fractions, and a random Pauli channel is used to study noise-dependent quantum bit error rate (QBER). Across 120 attack trials and 70 noise trials, the measured means broadly follow the ideal theoretical expectations. The work demonstrates protocol behavior in simulation; it is not a complete security proof or a production QKD implementation.

1. Objectives and protocol models

BB84. Alice randomly chooses a bit and an encoding basis (Z or X). Bob independently chooses a measurement basis. They keep only positions where their bases match, a process called basis sifting.

BBM92. A source prepares entangled Bell pairs shared between Alice and Bob. Each party measures in a randomly selected Z or X basis, then they retain matching-basis results. The simulated Bell state is |Phi+> = (|00> + |11>)/sqrt(2), which gives correlated outcomes in both matching bases.

Intercept-and-resend Eve. Eve attacks a fraction f of transmissions, measures in a randomly selected basis, and resends a state based on her result. In the ideal model, the expected sifted-key error rate is QBER = f/4.

Pauli channel. With probability p, the channel applies one of X, Y, or Z, selected uniformly. For this model, the expected QBER is 2p/3. The noise experiment estimates QBER from a randomly revealed sample and compares the estimate with an illustrative 11% threshold.

2. Experimental setup

Each attack setting was run 10 times with 2,000 transmissions per trial. The attack sweep used f = 0, 0.2, 0.4, 0.6, 0.8, and 1.0. The noise sweep used p = 0, 0.025, 0.05, 0.075, 0.10, 0.125, and 0.15, also with 10 trials per setting. Explicit random seeds were recorded in the CSV output. Error bars in the supplied figures represent standard deviation across trials (mean ± SD), not standard error or a confidence interval.

3. Intercept-and-resend results

Eve fraction	BB84 mean QBER	BBM92 mean QBER	Theory f/4
0%	0.00%	0.00%	0.00%
20%	4.86%	5.34%	5.00%
40%	10.06%	10.10%	10.00%
60%	15.03%	14.60%	15.00%
80%	20.81%	19.77%	20.00%
100%	25.55%	24.91%	25.00%

At zero attack, both simulations measured 0% QBER. At full attack, BB84 measured a mean QBER of 25.55%, compared with the 25% ideal prediction; BBM92 measured 24.91%. The means increase approximately linearly with attack fraction. The average sifted-key rates are close to 50%, consistent with independent random choices between two bases.

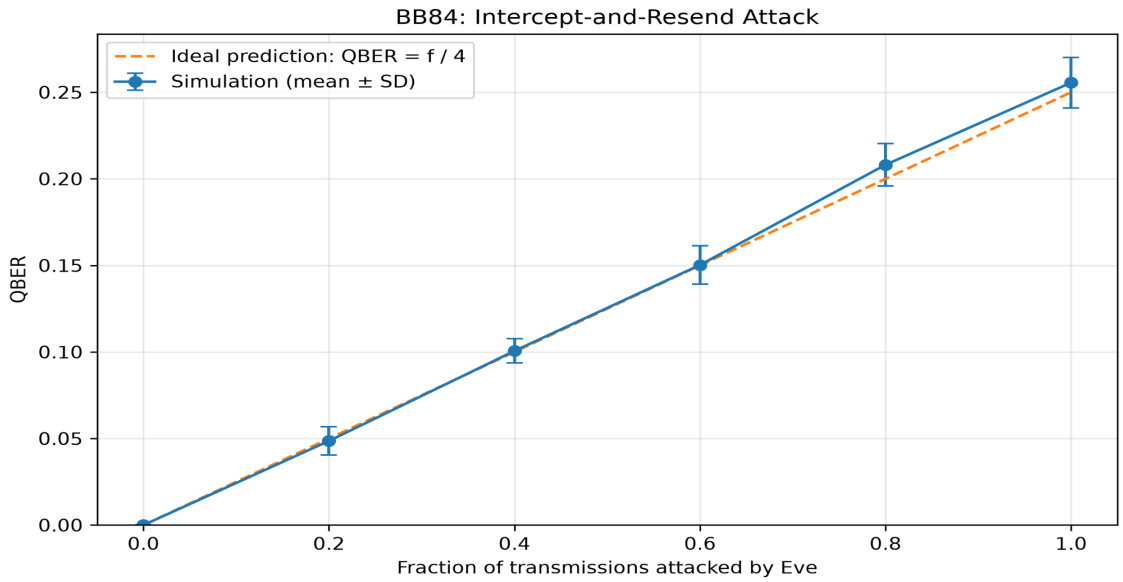


Figure 1. BB84 measured QBER versus Eve attack fraction.

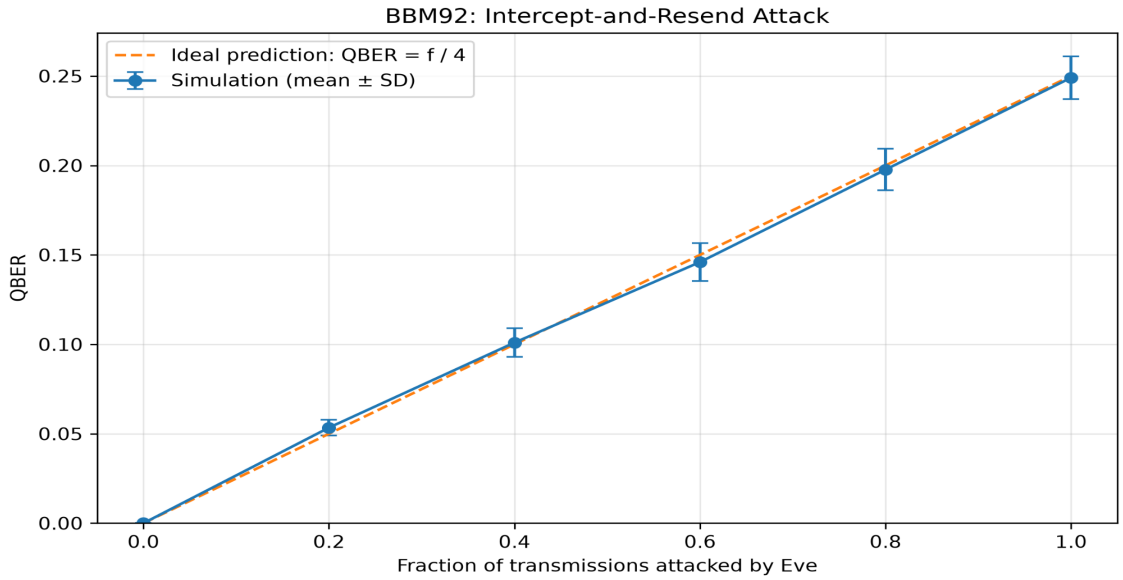


Figure 2. BBM92 measured QBER versus Eve attack fraction.

4. Channel-noise and parameter-estimation results

The measured sample QBER increases with the Pauli-error probability and broadly follows the theoretical expectation QBER = 2p/3. At p = 0.15, the mean sample QBER was 11.10%, compared with the ideal expectation of 10.00%. Six of ten trials exceeded the illustrative 11% abort threshold. No trials exceeded the threshold at the lower tested noise levels.

Pauli error p	Mean sample QBER	Theory 2p/3	Aborts / 10
0.000	0.00%	0.00%	0/10
0.025	1.44%	1.67%	0/10
0.050	3.09%	3.33%	0/10
0.075	5.09%	5.00%	0/10
0.100	6.86%	6.67%	0/10
0.125	8.44%	8.33%	0/10
0.150	11.10%	10.00%	6/10

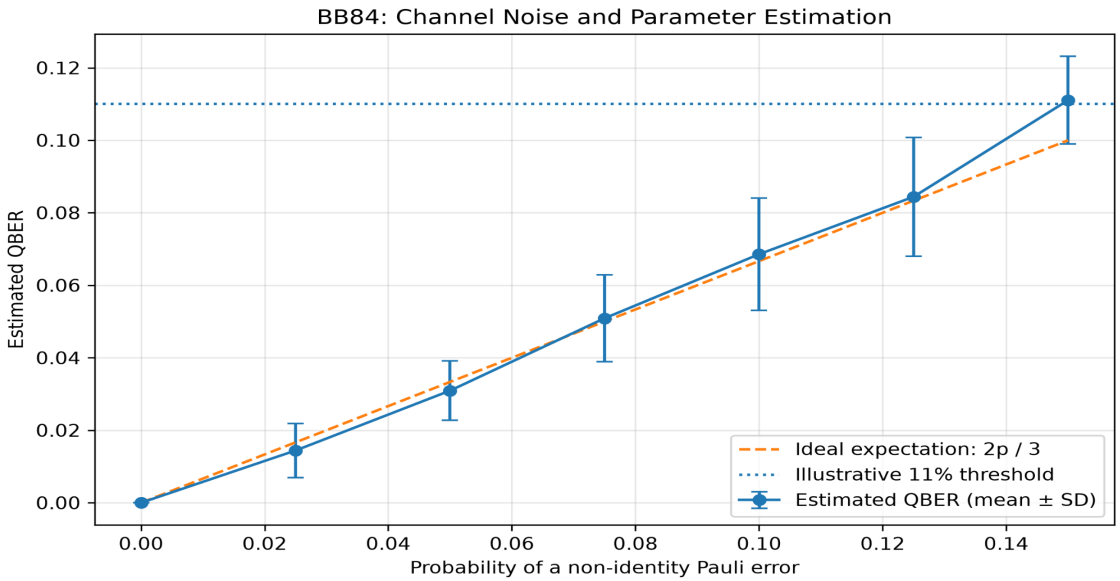


Figure 3. Sample QBER versus Pauli-error probability. Error bars show mean ± SD across 10 trials; the dotted line is the illustrative 11% decision threshold.

5. Verification and reproducibility

The project test suite reports 12 passed tests. The tests cover ideal BB84 key agreement, approximate basis-sifting rate, full and zero Eve attack cases, ideal BBM92 behavior, and noise-model input validation and threshold behavior. The experiment runner saves per-trial measurements, parameters, and random seeds to CSV and saves comparison figures to PNG.

6. Limitations and security interpretation

The 11% threshold is an illustrative rule used for demonstration; it is not a universal security threshold and does not by itself establish secrecy. A complete QKD implementation needs a protocol-specific security analysis, finite-key parameter estimation, authenticated classical communication, error correction, privacy amplification, and careful handling of information revealed during testing. The current simulation does not implement complete error correction or privacy amplification, and a candidate key whose bits disagree is not ready for cryptographic use. Results are from a simulator, not a physical quantum channel or hardware experiment.

7. Conclusion

The experiments reproduce two expected ideal-model trends: intercept-and-resend QBER grows approximately as f/4 for both BB84 and BBM92, and the random Pauli channel produces QBER that grows approximately as 2p/3. The measured means agree closely with theory within trial-to-trial variation. The work provides a reproducible educational comparison of

prepare-and-measure and entanglement-based QKD, while leaving full key distillation and rigorous security analysis as future work.

Reproducibility artifacts: bb84_eve_results.csv, bbm92_eve_results.csv, channel_noise_results.csv, and the three comparison PNG figures generated by scripts/run_experiments.py.